



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Resecurity Details Anubis Ransomware Attack on Adriatic Port Authority

Malware Analysis

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Resecurity Details Anubis Ransomware Attack on Adriatic Port Authority

1. Executive Summary

Lost Boys Cyber assesses with medium confidence that the Resecurity-reported Anubis ransomware intrusion against the Adriatic Port Authority is best treated as a source-backed ransomware case study for maritime critical-infrastructure defenders rather than a sample-backed reverse-engineering report. The available reporting describes a targeted intrusion against the Port System Authority of the Central Adriatic Sea / Port of Ancona ecosystem in which Anubis operators allegedly used spear phishing, privilege escalation, exploitation of unpatched IT vulnerabilities, and double-extortion pressure to disrupt port-adjacent business systems.

The incident is operationally important because it demonstrates how ransomware activity that remains within enterprise IT can still produce cyber-physical consequences for maritime logistics. Reporting states that cargo tracking, shipping schedules, customs-processing functions, partner confidence, and regional logistics operations were affected, while sensitive contracts, employee records, safety plans, and security-operations information were exposed. This should be read as a warning about IT-to-business-process dependency in ports, not as evidence that Anubis directly manipulated OT controllers, vessel systems, cranes, or industrial safety equipment.

No original Anubis binary, ransom note image, leak-site dataset, packet capture, forensic image, or victim telemetry was provided to Lost Boys Cyber for independent static analysis, code analysis, or detonation. The malware-analysis sections below therefore separate reported ransomware tradecraft from analyst-derived detection, hunting, and response guidance. Where the report discusses encryption, exfiltration, RaaS operations, or maritime disruption, the basis is Resecurity and Industrial Cyber reporting plus public contextual reporting from Ports Europe.

Immediate defensive priorities for maritime, logistics, and port-authority environments are to harden identity and remote-access controls, require phishing-resistant MFA on Microsoft 365 / Azure and VPN access, patch internet-facing VPN and web applications, segment business-critical logistics applications from user workstations, validate immutable backups, monitor for mass file encryption and unusual archive/exfiltration staging, and rehearse port-specific ransomware continuity procedures before a disruption forces manual cargo workflows.

Finding	Analyst Assessment	Defensive Implication
Spear phishing reported as initial access	Medium confidence; reported by Resecurity, not independently validated by Lost Boys Cyber	Prioritize attachment detonation, mail authentication, user-reporting workflows, and conditional-access enforcement
IT vulnerabilities and insecure cloud/admin accounts reportedly enabled spread	Medium-to-high confidence as a common ransomware pattern and explicitly cited in source reporting	Patch VPN/web applications, remove stale privileged access, and require MFA for Microsoft 365/Azure administration
Data theft plus encryption used for pressure	High confidence from source reporting	Treat cases as data-breach and business-continuity incidents, not only malware cleanup
Port logistics disruption occurred without reported OT compromise	Medium confidence	Map business-process dependencies and segment critical logistics applications from broad corporate access
Safety plans and security-operations data reportedly targeted	Medium confidence	Review leaked-sensitive-document exposure as a physical-security, insider-threat, and smuggling-risk concern

2. Sample Metadata

Field	Value
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Family	Anubis ransomware
Malware Type	Ransomware / double-extortion intrusion set; source-backed case study
Platform	Enterprise IT networks; likely Windows-centric ransomware operations, but no binary was provided for platform validation
Primary Sample SHA256	Not available in reviewed reporting
Additional Hashes	No sample hashes were published in the reviewed sources
File Type / Format	Unknown; no executable, script, loader, archive, or ransom-note file was provided to Lost Boys Cyber
Delivery Theme	Reported spear-phishing email with malicious attachment targeting port-authority employees
First Seen / Reported	Compromise initially reported as occurring in December 2025; Anubis public claim and leak reporting followed in January 2026; Resecurity/Industrial Cyber reporting reviewed on 2026-06-17
Analyst Confidence	Medium overall; high for the fact that public reporting describes an Anubis ransomware/extortion event; low for malware internals because no sample was available
Primary Objective	File encryption, data theft, extortion, and disruption of maritime logistics business processes
Source Basis	Resecurity primary blog, Industrial Cyber summary, and Ports Europe contextual reporting

3. Source Review & Confidence

Source	Contribution	Confidence	Notes
Resecurity, “The Anubis Ransomware Attack on the Adriatic Port Authority”	Primary source for reported attack flow, Anubis background, ransom demand, operational impact, and mitigation themes	Medium	Detailed narrative but limited public artefacts; no hashes, ransom-note text, packet captures, or forensic indicators in the reviewed public text
Industrial Cyber, “Resecurity details Anubis ransomware attack on Adriatic Port Authority...”	Secondary sector-focused summary of the Resecurity findings and maritime implications	Medium	Useful for critical-infrastructure framing and source corroboration; mostly restates Resecurity claims
Ports Europe, “Ancona port hit by cyber attack with 56,000 files leaked”	Contextual reporting that the port authority confirmed a cyberattack and that approximately 56,000 internal files were reportedly stolen/published	Medium	Article is partially paywalled but accessible snippet corroborates public incident awareness and leak scale
Lost Boys Cyber analytic synthesis	Detection engineering, threat hunting, incident response, and	Medium	Derived from source facts and standard ransomware telemetry

	maritime resilience assessment		patterns; not sample-backed
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Directly reported evidence includes: Anubis targeting of the Adriatic Port Authority / Port of Ancona ecosystem, suspected spear-phishing initial access, exploitation of unpatched IT vulnerabilities, privilege escalation and lateral movement, file encryption affecting logistics applications, exfiltration of contracts and employee records, a reported US\$10 million Bitcoin ransom demand, a seven-day leak threat, and operational disruption that forced shipment delays or vessel rerouting.

High-confidence assessment: even if OT controllers were not directly targeted, ransomware against port-authority IT can produce operational consequences when cargo tracking, customs processing, scheduling, and partner coordination depend on compromised digital services.

Lower-confidence or unverified areas: the exact initial-access attachment, malware build, encryption implementation, affiliate identity, infrastructure, command-line behavior, leak-site authenticity, and victim-side process tree cannot be independently confirmed from the public material reviewed by Lost Boys Cyber.

4. Initial Triage and Delivery Assessment

The publicly reported intrusion began with a spear-phishing email sent to employees of the company managing the port authority. Resecurity states that the message contained a malicious attachment which, once opened, installed or launched ransomware within the organization's network. The source narrative does not publish the attachment name, file type, hash, command line, exploit chain, or mailbox telemetry; therefore, defenders should treat the delivery vector as a reported intrusion lead rather than a reusable sample signature.

After initial access, the attackers reportedly escalated privileges and moved laterally through the IT environment by exploiting unpatched vulnerabilities and weak or insecure accounts, including accounts associated with Office 365 / Azure administration. This pattern is consistent with ransomware intrusions in which phishing or brokered access is only the entry point and the damaging activity depends on credential abuse, remote administration, unmanaged privileged accounts, and insufficient segmentation.

Variant / Artefact	Delivery Role	Notes
Spear-phishing email	Reported initial access	Source states employees were targeted as a privileged-access weak link; specific subject, sender, and attachment metadata are not public
Malicious attachment	Payload launch or installer	File type and hash not available; do not build hash-only detection assumptions around this report
Compromised credentials / cloud accounts	Lateral access enabler	Resecurity cites insecure Office 365 / Azure account management as an example IT weakness producing cyber-physical effects
Unpatched IT vulnerabilities	Privilege escalation / lateral movement	Source mentions exploitation of unpatched infrastructure but does not list CVEs for the victim environment
Ransomware encryptor	Disruption component	Encrypted thousands of files and affected cargo tracking, schedules, and customs processing according to source reporting
Exfiltration staging	Double-extortion pressure	Contracts, employee records, safety plans, and security-operation information were reportedly stolen

5. Static Analysis

5.1. File and Build Characteristics

No malware sample was provided for hashing, PE header inspection, entropy analysis, packer identification, import analysis, timestamp review, section review, or compiler-signature assessment. Static analysis for this report is therefore limited to source-reported artefact categories and defender-facing assumptions that should be validated if a sample or victim telemetry becomes available.

Static Signal	Evidence	Analyst Value
Ransomware family label	Resecurity and Industrial Cyber identify the actor/tooling as Anubis ransomware	Establishes the case-study family but does not prove a specific binary lineage
No public sample hash	Reviewed sources do not provide MD5/SHA1/SHA256 values	Prevents reliable hash-based blocking; behavior and identity controls matter more
Double-extortion pattern	Reported encryption plus data theft and leak threat	Indicates intrusion scoping must include exfiltration, legal, privacy, and partner-notification workflows
IT-only compromise with operational effect	Resecurity states no specific OT targeting was required	Helps prioritize logistics application resilience and identity segmentation over narrow PLC-only monitoring
Safety/security document theft	Source says safety plans and security-operations information were targeted	Raises physical-security, smuggling, insider-recruitment, and facility-security implications

5.2. Strings, Imports, and Embedded Artefacts

No strings, imports, mutexes, resources, embedded URLs, ransom-note filenames, registry paths, service names, or compiler artefacts are publicly available in the reviewed reporting. If responders obtain a sample or victim host image, the first-pass static collection should preserve:

Artefact Class	Collection Goal	Why It Matters
Ransom note text and filename	Identify contact channels, negotiation identifiers, leak-site URLs, wallet references, and victim IDs	Supports family clustering and legal/notification workflows
Encryptor imports and strings	Identify file-walk logic, cryptographic APIs, shadow-copy deletion, service control, and process-kill lists	Helps distinguish destructive encryptor behavior from generic malware
Embedded configuration	Recover public keys, campaign IDs, extension markers, allow/deny path lists, and operator endpoints	Useful for scoping and controlled emulation
Packer/obfuscation markers	Determine whether unpacking is needed before static signatures are reliable	Prevents false confidence from superficial strings output
Exfiltration tooling references	Search for Rclone, Mega, cloud storage, FTP/SFTP, archivers, and temporary staging directories	Supports double-extortion scoping and log-retention priorities

5.3. Static Configuration Clues

Because no binary configuration is available, this report cannot identify Anubis-specific public keys, excluded directories, extension changes, mutexes, process-kill lists, or C2 endpoints for the Adriatic Port Authority case. The most useful configuration-level clues are operational rather than binary: a reported US\$10 million ransom demand, a seven-day leak deadline, claimed exfiltration of contracts and employee/safety/security data, and public attribution to an Anubis RaaS affiliate ecosystem active since at least late 2024 or early 2025.

6. Code Analysis

6.1. Reverse-Engineering Method

Lost Boys Cyber did not receive executable malware or forensic artefacts for Ghidra, radare2, strings, FLOSS, capa, YARA, sandbox detonation, or debugger-assisted review. Code-analysis judgments in this report are therefore explicitly source-backed. If a sample becomes available, analysis should focus on encryption implementation, privilege requirements, process termination, backup destruction, file-extension markers, ransom-note generation, logging, excluded paths, and any network or exfiltration modules.

6.2. Functions and Logic of Interest

Function / Routine	Evidence	Why It Matters
Initial access execution	Resecurity reports a malicious attachment delivered through spear phishing	Determines whether controls should focus on document macros, LNK/ISO/JS, archive content, or executable attachments once artefacts are recovered
Privilege escalation	Source reporting states attackers escalated privileges after access	Privileged identity compromise is often the gate between one infected workstation and enterprise encryption
Lateral movement	Source reporting states unpatched vulnerabilities and IT weaknesses were exploited	Drives hunting across VPN, RDP, SMB, WinRM, cloud admin, EDR tamper, and remote-management logs
Data exfiltration	Contracts, employee records, safety plans, and security-operation information were reportedly stolen	Requires breach scoping and data-protection response beyond decrypt/restore operations
File encryption	Thousands of files were reportedly encrypted	Requires rapid containment, backup validation, and detection of mass file modification
Extortion communications	Reported US\$10 million Bitcoin demand with seven-day leak threat	Creates time pressure for legal, executive, insurance, and regulator coordination

6.3. Anti-Analysis, Evasion, and Obfuscation

The reviewed public reporting does not publish Anubis sample internals for this incident. However, Resecurity describes Anubis as adaptive and multi-stage, with ecosystem characteristics including social engineering, credential theft, exploitation of VPN and web-application vulnerabilities, account takeover, and advanced defense evasion including QEMU virtual-machine abuse in broader Anubis operations. Defenders should not assume a simple single-stage encryptor. Investigations should preserve logs for the full intrusion chain, including initial access, staging, tool transfer, EDR tampering, backup targeting, exfiltration, and encryption.

Evasion / Tradecraft Area	Source Basis	Defensive Review Point
Social engineering	Reported spear-phishing attachment	Mailbox telemetry, attachment sandbox

		verdicts, user click/open timeline
Account takeover	Resecurity notes ATO and insecure cloud/admin accounts in Anubis operations	Conditional access, OAuth/app-consent events, Azure sign-in risk, impossible travel, MFA resets
Public-facing application exploitation	Resecurity references Anubis use of VPN and web app N-days / CVE derivatives	Patch status, exploit attempts, web logs, anomalous admin sessions
Defense evasion	Resecurity describes advanced evasion in broader Anubis operations	EDR tamper alerts, service stops, tool exclusions, virtualization/sandbox bypass indicators
Double extortion	Source reports exfiltration and leak threats	Outbound data transfer, archiver use, staging directories, cloud-storage access

7. Dynamic Analysis

7.1. Execution Overview

No detonation was performed because no sample was provided. The source-backed intrusion sequence is: phishing delivery to port-authority employees; malicious attachment execution; privilege escalation and lateral movement through vulnerable or weakly protected IT systems; access to systems supporting cargo tracking, shipping schedules, and customs processing; data exfiltration; ransomware encryption; ransom demand and leak threat; containment and restoration using backups.

This sequence should be validated against victim telemetry before operational conclusions are finalized. In a live response, the highest-value artefacts would be EDR process timelines, Windows Security logs, Azure/Microsoft 365 sign-ins, VPN authentication logs, web/VPN exploit traces, file-server mass-write events, backup-system activity, data-transfer logs, and ransom-note creation timestamps.

7.2. Filesystem, Registry, Service, and Task Changes

Artefact / Telemetry	Reported or Expected Behavior	Defensive Value
File-server mass modification	Source reports thousands of encrypted files	Detect rapid extension changes, high-volume writes, and entropy changes from a single principal
Ransom notes	Expected for ransomware operations but not published in reviewed text	Establishes encryption time window, affiliate ID, and extortion channels
Archive staging directories	Expected for double extortion; specific paths unavailable	Hunt for large .7z/.zip/.rar creation before encryption
Backup-system interaction	Recovery was reportedly slowed by outdated backup protocols	Review deletion/tamper attempts and backup immutability gaps
Security tool tamper events	Common ransomware precursor; not specifically reported for the victim	Preserve EDR/AV service stop, exclusion, and policy-change events
Cloud/admin account changes	Resecurity cites insecure Office 365 / Azure accounts as an IT weakness example	Review admin role assignment, mailbox export, OAuth grant, and conditional-access changes

7.3. Network and Command-and-Control Behaviour

No Anubis C2 domains, IP addresses, leak-site URLs, wallets, or malware download endpoints were published in the reviewed sources. Network analysis should therefore focus on behavior classes rather than brittle IOCs: anomalous outbound bulk transfer, new cloud-storage use, remote-access sessions from unusual infrastructure, VPN/web exploitation attempts,

administrative protocol fan-out, and connections to known ransomware leak or negotiation infrastructure if available from victim-side evidence.

Network Signal	Role	Stability / Confidence	Notes
Unusual Microsoft 365 / Azure administrative activity	Identity compromise / cloud foothold	Medium / source-informed	Resecurity cites insecure cloud/admin accounts as a relevant weakness; validate with tenant logs
VPN or web application exploit traces	Initial access or lateral movement	Medium / source-informed	Source references Anubis exploitation of public-facing apps and unpatched IT vulnerabilities
Large outbound transfers before encryption	Data exfiltration	High / expected from double extortion	Tune per environment; correlate with archive creation and new external destinations
RDP/SMB/WinRM fan-out	Lateral movement	Medium / expected	Look for one account touching many servers shortly before encryption
Leak-site or negotiation infrastructure	Extortion	Low from public reporting	Specific URLs not provided; obtain from ransom note or victim comms if available

7.4. Observed Capability Summary

Capability	Supporting Dynamic Evidence
Initial access through phishing	Resecurity states employees received spear-phishing email with malicious attachment
Privilege escalation and lateral movement	Source reports attackers escalated privileges and exploited unpatched IT vulnerabilities
File encryption	Source reports thousands of files encrypted and logistics systems rendered inaccessible
Data theft	Source reports exfiltration of contracts, employee records, safety plans, and security-operation information
Double extortion	Source reports US\$10 million Bitcoin demand and seven-day leak threat
Business-process disruption	Source reports cargo tracking, shipping schedules, customs processing, rerouting, and shipment disruption

8. Process Tree and Execution Chain

No victim EDR process tree was available. The following execution chain is a source-backed incident model intended to guide telemetry review and detection engineering. Analysts should replace the generic parent/child placeholders with real process, account, and host data if forensic artefacts become available.

Step	Parent Process	Child / Action	Why It Matters
1	User mailbox / email client	Opens spear-phishing	Establishes patient-zero user,

		attachment	delivery time, attachment hash, and mail-flow controls
2	Unknown attachment process	Executes downloader, script, or ransomware stage	Missing artefact; recover from EDR command lines and prefetch/Amcache/Sysmon if available
3	Compromised user context	Credential theft, privilege escalation, or admin session creation	Key pivot from endpoint infection to enterprise compromise
4	Admin or service account	Lateral movement to file servers, application servers, or cloud admin portals	Drives blast radius and identifies compromised identity material
5	Operator tooling / ransomware process	Exfiltration staging and outbound transfer	Data-breach scoping and legal/regulatory obligations depend on this phase
6	Ransomware encryptor	Mass file writes, ransom-note creation, service interruption	Highest-confidence disruption phase and primary containment trigger
7	Recovery and containment tooling	Isolation, forensic collection, backup restore	Determines whether reinfection risk and root cause are adequately addressed

9. Persistence and System Modification

The public sources do not document a specific Anubis persistence mechanism used inside the Adriatic Port Authority environment. The intrusion reportedly progressed far enough to encrypt enterprise data, which implies the operators achieved durable enough access for lateral movement and staging even if formal persistence was not disclosed.

Modification Type	Artefact	Evidence	Confidence
Privileged account abuse	Office 365 / Azure or other administrative accounts	Resecurity cites insecure accounts managing Office 365 / Azure as an example IT weakness producing impact	Medium
Vulnerable service exploitation	VPN, web app, or public-facing IT infrastructure	Resecurity describes Anubis use of public-facing application exploitation and unpatched vulnerabilities	Medium
Backup and recovery impact	Outdated backup protocols slowed restoration	Resecurity reports backup limitations in recovery	Medium
Ransomware encryption state	Encrypted business files and inaccessible logistics systems	Source reporting states thousands of files were encrypted	High
Data-theft residue	Contracts, employee records, safety/security operations information	Source reporting states these data categories were exfiltrated	Medium

Defenders should review persistence through scheduled tasks, services, WMI subscriptions, remote-management agents, VPN accounts, cloud application consent grants, mailbox forwarding rules, privileged role assignments, identity-provider policy changes, and backup administrator access. Ransomware response should not declare eradication complete until identity persistence and cloud control-plane persistence have been checked.

10. Network and Infrastructure Analysis

The reviewed public reporting does not expose stable infrastructure IOCs for the Adriatic Port Authority Anubis case. Resecurity's broader Anubis overview states the group has recruited affiliates through underground forums, advertised data-extortion and initial-access-broker monetization, and sought compromised remote access to enterprises in the United States, Canada, the European Union, and Australia while avoiding ex-USSR and BRICS countries. That operating model suggests infrastructure may vary by affiliate and victim.

IOC / Infrastructure	Type	Role	Stability / Confidence
Anubis leak/extortion presence	Actor-controlled public pressure channel	Publication or threatened publication of stolen data	Medium; specific URL not included in reviewed source
RAMP / XSS / Exploit forum presence	Cybercrime forum ecosystem	Affiliate recruitment, RaaS promotion, data purchase, initial-access sourcing	Medium; source-reported broader ecosystem context
Office 365 / Azure admin interfaces	Cloud control plane	Identity and administrative access abuse risk	Medium; source-cited weakness example, not a victim IOC
VPN and web-application attack surface	Public-facing access layer	Potential initial access and lateral foothold	Medium; source describes Anubis use of public-facing app exploitation generally
Unknown exfiltration destination	External network or cloud storage	Transfer of contracts, employee data, safety/security documents	High for exfiltration occurrence; low for specific infrastructure

Because hard infrastructure is missing, defensive use should prioritize telemetry correlations: login origin anomalies, impossible travel, newly observed autonomous systems, abnormal cloud storage, large encrypted archives leaving the network, RDP/VPN sessions outside business patterns, and repeated authentication to maritime logistics applications by newly privileged accounts.

11. Indicators of Compromise

This report intentionally avoids inventing file hashes or network IOCs not present in the reviewed reporting. Machine-readable indicators are limited to source URLs and source-described observable categories. Operational detection should be behavior-led until victim artefacts or a verified Anubis sample are obtained.

IOC Type	Example / Count	Source	Operational Notes
Malware hashes	0 published	Reviewed sources	Do not rely on hash blocking for this case
Domains / IPs	0 victim-specific malware infrastructure items published	Reviewed sources	Monitor for behavior and obtain ransom-note/leak-site artefacts from IR team
Source URLs	3 report/source URLs	Resecurity, Industrial Cyber, Ports Europe	Useful for evidence traceability, not network blocking
Actor/family labels	Anubis ransomware;	Resecurity	Labels help threat tracking but

	Anubis__media; superSonic		are not technical IOCs
Victim/targeting descriptors	Adriatic Port Authority; Port of Ancona; maritime logistics	Public reporting	Useful for sector risk modeling and executive context
Data categories	Contracts, employee records, safety plans, security-operations information	Resecurity / Industrial Cyber	Drives exposure review and partner notification

12. MITRE ATT&CK Mapping

Technique	Name	Evidence
T1566.001	Phishing: Spearphishing Attachment	Resecurity reports spear-phishing email with malicious attachment as suspected initial access
T1190	Exploit Public-Facing Application	Resecurity describes Anubis exploitation of public-facing applications and unpatched IT vulnerabilities; victim-specific CVEs are not listed
T1078	Valid Accounts	Source references insecure Office 365 / Azure accounts and account takeover patterns in Anubis operations
T1068	Exploitation for Privilege Escalation	Attackers reportedly escalated privileges after initial access
T1021	Remote Services	Lateral movement through IT environment is reported; exact protocol unknown
T1083	File and Directory Discovery	Ransomware operators likely enumerated data stores before encryption/exfiltration; inferred from successful targeting of contracts, employee records, and safety documents
T1560	Archive Collected Data	Inferred likely staging behavior for double extortion; not directly confirmed in public text
T1041	Exfiltration Over C2 Channel	Data exfiltration is reported; mechanism not disclosed
T1486	Data Encrypted for Impact	Source reports thousands of files encrypted and logistics functions disrupted
T1490	Inhibit System Recovery	Backup recovery was reportedly slowed; direct deletion of backups is not confirmed, but recovery infrastructure should be reviewed
T1657	Financial Theft	Ransom payment demand of US\$10 million in Bitcoin is reported

13. Detection Engineering

13.1. Detection Logic Summary

Signal	Telemetry Source	Analytic Goal	Tuning Notes
Spear-phishing attachment execution followed by admin activity	Email gateway, EDR, identity logs	Identify user-to-admin pivot shortly after suspicious attachment open	Tune by attachment sandbox verdict, sender reputation, and new admin role/session creation
Mass file encryption on file shares	EDR, Sysmon, Windows file server auditing, storage telemetry	Detect rapid extension changes, entropy increases, or ransom-note fan-out	Exclude known backup/indexing jobs; alert on new process/account baselines
Archive staging before outbound transfer	EDR process and file events	Catch data theft preceding ransomware encryption	Tune for approved backup/compression jobs and expected export workflows
Suspicious cloud/admin account changes	Azure AD / Entra ID, Microsoft 365 audit logs	Detect identity persistence and privilege escalation	Prioritize privileged role assignment, MFA reset, conditional-access edits, OAuth grants
RDP/SMB/WinRM fan-out from one account	Windows Security, EDR, network telemetry	Identify lateral movement across port logistics servers	Tune for management servers, vulnerability scanners, and patch tools
Security tool tampering or backup access anomalies	EDR, service logs, backup platform logs	Detect preparation for enterprise encryption	Treat as high severity when correlated with mass file writes or new privileged sessions

13.2. Detection Content

```

title: Possible Ransomware Mass File Modification on Windows File Shares
id: 6d8d6582-78fe-42ec-bf5d-anubis-port-ransomware-behavior
status: experimental
description: Detects high-volume file modification and ransom-note style writes consistent with
ransomware impact. Tune thresholds per file-server baseline.
author: Lost Boys Cyber
references:
  - https://www.resecurity.com/blog/article/the-anubis-ransomware-attack-on-the-adriatic-port-authority
logsource:
  product: windows
  category: file_event
detection:
  selection_note:
    TargetFilename|contains:
      - 'README'
      - 'RECOVER'
      - 'RESTORE'
      - 'DECRYPT'
      - 'RANSOM'
    TargetFilename|endswith:
      - '.txt'
      - '.html'
      - '.hta'
  selection_mass_modify:
    EventID:
      - 11
      - 15
timeframe: 10m
condition: selection_note or selection_mass_modify

```

```

fields:
  - Computer
  - User
  - Image
  - TargetFilename
  - EventID
falsepositives:
  - Backup software, document-management migrations, legitimate mass rename jobs
level: high

title: Suspicious Archive Staging Followed by External Transfer Tools
id: 1f5743be-8df7-4d48-a41c-anubis-double-extortion-staging
status: experimental
description: Detects common archive or transfer utilities that may support ransomware double-extortion staging before encryption.
author: Lost Boys Cyber
logsource:
  product: windows
  category: process_creation
detection:
  archive_tools:
    Image|endswith:
      - '\\7z.exe'
      - '\\7za.exe'
      - '\\rar.exe'
      - '\\winrar.exe'
      - '\\tar.exe'
      - '\\powershell.exe'
  archive_args:
    CommandLine|contains:
      - ' a '
      - ' -mx'
      - ' Compress-Archive'
      - '.zip'
      - '.7z'
      - '.rar'
  transfer_tools:
    Image|endswith:
      - '\\rclone.exe'
      - '\\winscp.com'
      - '\\curl.exe'
      - '\\pscp.exe'
      - '\\ftp.exe'
  condition: (archive_tools and archive_args) or transfer_tools
fields:
  - UtcTime
  - Computer
  - User
  - ParentImage
  - Image
  - CommandLine
falsepositives:
  - Authorized backups, administrator data exports, software deployment jobs
level: medium

// Microsoft Defender XDR: mass file modification and ransom-note indicators
let lookback = 7d;
let ransomTerms = dynamic(['recover','restore','decrypt','ransom','readme']);
DeviceFileEvents
| where Timestamp > ago(lookback)
| where ActionType in ('FileCreated','FileModified','FileRenamed')
| extend lowerName = tolower(Filename)
| summarize FileEvents=count(), ExampleFiles=make_set(Filename, 10), Folders=make_set(FolderPath, 10)
  by DeviceName, InitiatingProcessAccountName, InitiatingProcessFileName, bin(Timestamp, 10m)
| where FileEvents > 500 or array_length(set_intersect(ExampleFiles, ransomTerms)) > 0
| order by FileEvents desc

// Microsoft Defender XDR: possible double-extortion staging through archive and transfer tooling

```

```

let lookback = 14d;
DeviceProcessEvents
| where Timestamp > ago(lookback)
| where FileName in~
('7z.exe', '7za.exe', 'rar.exe', 'winrar.exe', 'tar.exe', 'rclone.exe', 'winscp.com', 'curl.exe', 'pscp.exe', 'ftp.exe', 'powershell.exe')
| where ProcessCommandLine has_any ('.7z', '.zip', '.rar', 'Compress-Archive', 'mega', 'sftp', 'ftp', 'copyto', 'sync', '--transfers')
| project Timestamp, DeviceName, AccountName, InitiatingProcessFileName, FileName, ProcessCommandLine
| order by Timestamp desc

index=wineventlog (sourcetype="WinEventLog:Security" OR sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational")
(EventCode=4688 OR EventCode=1)
(Image="*\\7z.exe" OR Image="*\\7za.exe" OR Image="*\\rar.exe" OR Image="*\\rclone.exe" OR Image="*\\winscp.com" OR Image="*\\curl.exe" OR New_Process_Name="*\\7z.exe" OR New_Process_Name="*\\rclone.exe")
| eval cmd=coalesce(CommandLine, Process_Command_Line)
| where like(lower(cmd), "%7z%") OR like(lower(cmd), "%zip%") OR like(lower(cmd), "%copyto%") OR like(lower(cmd), "%sync%") OR like(lower(cmd), "%sftp%")
| table _time host user parent_process_name Image New_Process_Name cmd

```

13.3. Detection Validation and Blind Spots

These detections are behavior-led because the reviewed public reporting does not include Anubis file hashes or infrastructure. Validate them against known backup, document-management, and file-migration workflows before enabling blocking. Blind spots include encrypted exfiltration through sanctioned cloud services, unmanaged endpoints without EDR, file servers without object-level auditing, and ransomware affiliates that use renamed tools or native APIs instead of common archivers. The highest-confidence alerting comes from correlation: suspicious attachment open, new privileged authentication, lateral movement fan-out, archive staging, external transfer, EDR/backup tamper, and mass file modification within the same incident window.

14. Threat Hunting

14.1. Hunt Hypotheses

- A user who opened a suspicious attachment later authenticated to cloud or VPN resources from unusual infrastructure, leading to privileged role use or lateral movement.
- Before encryption, an operator staged contracts, employee records, safety plans, or security-operation documents into compressed archives and transferred them externally.
- Ransomware impact was preceded by security-tool tampering, backup-console access, or service/process termination on file and application servers.
- Maritime logistics applications experienced authentication or service anomalies before business-process disruption became visible.

14.2. Hunt Queries

```

// Hunt: suspicious email-to-identity pivot after attachment delivery
let lookback = 14d;
let suspiciousRecipients = EmailEvents
| where Timestamp > ago(lookback)
| where ThreatTypes has_any ('Malware', 'Phish') or DeliveryAction in ('Delivered', 'Junked')
| project RecipientEmailAddress, EmailTime=Timestamp, Subject, SenderFromAddress, NetworkMessageId;
AADSignInEventsBeta
| where Timestamp > ago(lookback)
| join kind=inner suspiciousRecipients on $left.AccountUpn == $right.RecipientEmailAddress
| where Timestamp between (EmailTime .. EmailTime + 2d)
| where RiskLevelDuringSignIn != 'none' or ConditionalAccessStatus != 'success' or
isnotempty(IPAddress)
| project Timestamp, AccountUpn, IPAddress, Application, ResourceDisplayName, ConditionalAccessStatus,
RiskLevelDuringSignIn, EmailTime, Subject, SenderFromAddress
| order by Timestamp desc

// Hunt: high-volume archive creation in sensitive maritime/logistics document locations
let lookback = 30d;

```

```

DeviceFileEvents
| where Timestamp > ago(lookback)
| where FileName endswith '.zip' or FileName endswith '.7z' or FileName endswith '.rar'
| where FolderPath has_any
('contract','contracts','employee','safety','security','customs','cargo','shipping','schedule','port')
| summarize ArchiveCount=count(), Archives=make_set(FolderPath, 20) by DeviceName,
InitiatingProcessAccountName, InitiatingProcessFileName, bin(Timestamp, 1h)
| where ArchiveCount >= 5
| order by ArchiveCount desc

// Hunt: possible ransomware preparation through security or backup tampering
let lookback = 14d;
DeviceProcessEvents
| where Timestamp > ago(lookback)
| where ProcessCommandLine has_any ('vssadmin delete shadows','wmic shadowcopy delete','bcdedit
/set','wbadmin delete','wevtutil cl','sc stop','net stop','Set-MpPreference','Add-MpPreference')
| project Timestamp, DeviceName, AccountName, InitiatingProcessFileName, FileName, ProcessCommandLine
| order by Timestamp desc

index=azure sourcetype="azure:aad:signin" earliest=-14d
| search riskLevelDuringSignIn!=none OR conditionalAccessStatus!=success
| stats count min(_time) as first_seen max(_time) as last_seen values(appDisplayName) as apps
values(ipAddress) as ips by userPrincipalName
| where count > 3
| convert ctime(first_seen) ctime(last_seen)

```

14.3. Triage Guidance for Hunt Hits

A meaningful hunt hit is not a single archive command or failed sign-in; it is a sequence that connects initial access, identity abuse, lateral movement, data staging, and mass file modification. Prioritize hits involving privileged accounts, file servers, maritime logistics systems, customs/cargo applications, backup administrators, and users with access to safety or security-operation documents. Benign lookalikes include backup jobs, file migrations, approved records exports, eDiscovery, vulnerability scanners, and administrator maintenance. Escalate to incident response when a suspicious sequence spans multiple hosts, shows external transfer, or touches sensitive document repositories.

15. Triage and Incident Response

Treat an Anubis-like alert in a port or logistics environment as a potential enterprise intrusion with operational-consequence risk. Do not scope it as a single infected workstation until identity, file-server, cloud, backup, and logistics-application telemetry have been reviewed.

Priority	Action	Owner	Rationale
Critical	Isolate actively encrypting hosts and disable suspected compromised accounts	SOC / IT Operations	Stops further encryption and lateral movement
Critical	Preserve ransom notes, EDR timelines, cloud sign-ins, VPN logs, and file-server audit logs	Incident Response	Maintains evidence for root-cause, legal, and insurance decisions
High	Freeze and validate backups before restore	Backup / Infrastructure	Prevents overwriting clean backups or restoring into active compromise
High	Review exfiltration exposure for contracts, employee records, safety plans, and security operations	Legal / Privacy / Security	Source reporting shows these data classes may have physical and regulatory implications

High	Coordinate manual cargo, customs, and schedule continuity procedures	Operations / Port Authority	Maintains maritime business process while IT recovery proceeds
Medium	Notify trusted partners and sector ISAC channels as appropriate	Executive / Comms / Security	Supports supply-chain coordination without over-disclosing unverified details

Initial collection should include: patient-zero mailbox, suspicious attachment, endpoint triage package, Windows Security/Sysmon logs, EDR raw process tree, Azure/Entra ID audit and sign-in logs, Microsoft 365 unified audit logs, VPN/RDP logs, file-server change logs, backup-console logs, firewall/proxy records, ransom note, external transfer logs, and any leak-site communications.

16. Validation and Emulation

Validation completed for this report consisted of source retrieval, source comparison, and analytic review. Lost Boys Cyber did not detonate a sample, recover a binary configuration, decompile ransomware code, or validate victim-side process telemetry. The report should be refreshed if any of the following become available: malware hash, ransom note, leaked sample, victim EDR export, cloud audit logs, negotiation portal URL, wallet address, public leak-site post, or third-party forensic report.

Safe emulation can be performed without live malware by replaying detection-relevant behaviors in a lab: creating high-volume benign files, generating test ransom-note filenames, compressing synthetic sensitive directories, simulating Rclone/curl outbound command lines to controlled destinations, producing Azure privileged role changes in a test tenant, and executing backup-tamper command lines with harmless echo wrappers. Do not run unknown Anubis binaries on production networks.

Validation Item	Status	Notes
Source article review	Complete	Resecurity, Industrial Cyber, and Ports Europe reviewed
Static malware analysis	Not performed	No sample available
Code/decompiler review	Not performed	No binary or script available
Sandbox execution	Not performed	No sample available
Detection logic review	Complete	Behavioral Sigma, KQL, and SPL drafted for ransomware/double-extortion patterns
PDF/build QA	Pending until renderer execution	Run analyst-kit QA/build before portal delivery

17. Recommendations

Priority	Owner	Action	Rationale
Critical	Identity / Cloud Engineering	Enforce phishing-resistant MFA for administrators, disable legacy authentication, and monitor privileged role changes in Microsoft 365 / Azure	Source reporting highlights insecure cloud/admin accounts as a path from IT compromise to operational impact
Critical	Infrastructure / Network	Patch and externally validate VPN, web application, and remote-access services; remove unsupported systems from	Resecurity describes Anubis exploitation of public-facing applications and unpatched IT

		internet exposure	vulnerabilities
Critical	Backup / Recovery	Implement immutable/offline backups and run port-operation restore exercises covering cargo, customs, and scheduling systems	Recovery was reportedly slowed by outdated backup protocols; backups determine extortion leverage
High	SOC / Detection Engineering	Deploy behavior-based ransomware detections for mass file modification, archive staging, transfer tooling, security-tool tamper, and lateral movement fan-out	Public IOCs are sparse; behavior correlation is more resilient
High	Security Architecture	Segment maritime logistics applications, file shares, and backup management from broad user workstation access	IT-only compromise can still disrupt cyber-physical business processes
High	Email / Endpoint Security	Strengthen attachment detonation, block high-risk archive/script/LNK patterns, and train users on targeted port-authority lures	Spear phishing is the reported initial access vector
Medium	Governance / Risk	Map sensitive safety plans, security-operation documents, contracts, and employee records to access controls and DLP coverage	These data categories were reportedly targeted and can create physical-security exposure
Medium	Incident Response	Update ransomware playbooks to include maritime continuity, vessel/customer communications, regulator notification, and manual operations	Port incidents require business-process coordination beyond endpoint eradication
Medium	Threat Intelligence	Track Anubis RaaS affiliate activity, leak-site claims, and maritime-sector ransomware reporting through trusted sources	Anubis infrastructure and affiliate tooling may shift rapidly

18. References and Refresh Notes

- [1] Resecurity, “The Anubis Ransomware Attack on the Adriatic Port Authority,” accessed 2026-06-17. <https://www.resecurity.com/blog/article/the-anubis-ransomware-attack-on-the-adriatic-port-authority>
- [2] Industrial Cyber, “Resecurity details Anubis ransomware attack on Adriatic Port Authority, exposing maritime infrastructure risks,” published 2026-06-17, accessed 2026-06-17. <https://industrialcyber.co/transport/resecurity-details-anubis-ransomware-attack-on-adriatic-port-authority-exposing-maritime-infrastructure-risks/>
- [3] Ports Europe, “Ancona port hit by cyber attack with 56,000 files leaked,” published 2026-01-16, accessed 2026-06-17. <https://www.portseurope.com/ancona-port-hit-by-cyber-attack-with-56000-files-leaked/>

Refresh triggers: update this report if Resecurity or another primary source publishes hashes, ransom-note content, leak-site URLs, wallet addresses, victim telemetry, specific exploited CVEs, decryptor information, or confirmed operational impact

timelines. Re-score detection and hunting content when any technical IOC is added because the current report is intentionally behavior-led due to lack of sample artefacts.